👩‍🏫 Staff Guide · Small Engines Overhaul · Unit 3 4-Stroke Theory & Engine Parts (Level 2)

Level 2 = about 3rd-grade reading. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: “Week 2: 4 Stroke Theory PowerPoint (Small Engines Parts ID Reading) + Parts ID quiz, Engine Tools/Parts sheets, Briggs & Stratton disassembly/assembly video, Marine 2-stroke and 4-stroke video notes, iCADEMY Bronze 6 Engine Principles of Operation”)	**Level 2 piece**
4 Stroke Theory PowerPoint (13 slides) + Parts ID quiz	Deck + notes A, B + quiz; Parts ID lab packet
Engine Tools/Parts sheets	Parts ID lab packet + vocab cards
B&S disassembly/assembly video	Video guide (link only)
Marine 2-stroke/4-stroke video notes	Slides “2-Stroke vs 4-Stroke” + notes C
iCADEMY Bronze 6 Engine Principles	Our own slides on the 4 strokes (no STIHL content)
Class slides	Deck: 15 slides, Words to Know first, hold-up-1-2-3 checks, recap
His worksheets (grouped into lessons)	Guided notes: 11 items + key
Class quiz / notes check	Quiz: 10 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- Short lines, key words bold, word bank in every notes box
- Quiz: 3 choices, no “all of the above”
- Act out the 4 strokes: students move hands (suck, squeeze, bang, blow)
- Cutaway engine or clear model if the shop has one

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the “look for” part

4. What to grade

- Parts ID quiz: L1 points to 8 parts on a real engine or picture
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **iCADEMY Bronze 6**: STIHL iCADEMY is login-only; we wrote our own engine-principles content.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.